

1. Advantages of Java

Java is one of the most popular programming languages because of its many benefits. The main advantages are:

1. **Platform Independent** – Java programs can run on any operating system using the Java Virtual Machine (JVM). This is known as "**Write Once, Run Anywhere (WORA).**"
2. **Object-Oriented** – Java follows Object-Oriented Programming (OOP) concepts such as encapsulation, inheritance, polymorphism, and abstraction, making programs modular and reusable.
3. **Simple and Easy to Learn** – Java has a simple syntax and removes complex features like pointers, making it beginner-friendly.
4. **Secure** – Java provides built-in security features such as bytecode verification, exception handling, and automatic memory management.
5. **Robust** – Java is reliable because it has strong memory management, exception handling, and garbage collection.
6. **Multithreading Support** – Java allows multiple tasks to execute simultaneously, improving application performance.
7. **High Performance** – The Just-In-Time (JIT) compiler improves execution speed by compiling bytecode into machine code.
8. **Distributed** – Java supports distributed computing through networking APIs, making it suitable for internet-based applications.
9. **Portable** – Java programs can be easily moved from one system to another without modification.
10. **Rich Standard Library** – Java provides a large collection of built-in libraries for database connectivity, networking, GUI development, file handling, and more.

2. Java Program to Print Student Details

```
class Student {  
    public static void main(String[] args) {  
  
        String name = "Renu";  
        int rollNo = 101;  
        String department = "Computer Science";  
        int age = 20;  
        double percentage = 88.5;  
    }  
}
```

```
        System.out.println("Student Details");
        System.out.println("-----");
        System.out.println("Name      : " + name);
        System.out.println("Roll No   : " + rollNo);
        System.out.println("Department : " + department);
        System.out.println("Age       : " + age);
        System.out.println("Percentage : " + percentage + "%");
    }
}
```

Sample Output

```
Student Details
-----
Name      : Renu
Roll No   : 101
Department : Computer Science
Age       : 20
Percentage : 88.5%
```